Multimodal Remote Sensing Data Classification Based on Gaussian Mixture Variational Dynamic Fusion Network

Haoyu Wang, *Member, IEEE*, Xiaomin Liu, *Member, IEEE*, Zhenzhuang Qiao, Guoqing Wang, *Member, IEEE*, and Haotian Chen

***Abstract*— With the development of sensor technology, the rational use of multimodal data has become a research hotspot in the field of remote sensing. The multimodal fusion method can effectively improve the accuracy of remote sensing data classification by using the complementary information of different modalities. However, the existing multimodal fusion methods face many challenges, including difficulties in suppressing spectral noise, fully mining contextual information, and learning the strong adaptive fusion pattern. To address the above challenges, a Gaussian mixture variational dynamic fusion network (GM-VDFN) is proposed. First, a multimodal multiscale spatial graph is constructed, and the graph convolution is used to learn the multiscale features. In this process, a spatial topology constraint based on GM (STC-GM) is proposed, which suppresses spectral noise by constraining the topological consistency of the two modalities. Second, a multiscale dynamic graph aggregation module (MDGAM) is constructed, which can capture the shareable class identification information from multiscale features and mine personalized fusion patterns suitable for each sample. Finally, the evidence lower bound for the multimodal joint distribution is derived, and a multimodal variational autoencoder (M-VAE) is designed. Optimizing the evidence lower bound to model multimodal joint distributions, thereby learning the strong adaptive fusion pattern between modalities. Experimental results on four fusion datasets (Houston 2013, Trento, MUUFL, and Houston 2018) show that GM-VDFN achieved state-of-the-art performance in multimodal remote sensing data classification tasks.**

***Index Terms*— Classification, feature fusion, hyperspectral image (HSI), light detection and ranging (LiDAR), variational autoencoder.**

I. Introduction

With the rapid growth of the number of sensors, the properties and applications of various remote sensing data, such as hyperspectral images (HSIs) and light detection and ranging (LiDAR), have received extensive attention from researchers. HSIs have high spectral resolution [1], [2], [3], which can provide key information for distinguishing ground objects that are similar in appearance but significantly different in spectral [4]. The use of HSIs for ground object classification has shown promising applications in many fields, including environmental monitoring, ground object analysis, agricultural management, urban planning, and so on [5], [6]. However, it is difficult to distinguish ground objects with similar spectra using only HSIs [7]. As shown in Fig. 1(a), "Highway" and "Railway" have similar spectral curves, so it is easy to confuse the two classes by using only the spectral information of HSIs. The elevation information contained in LiDAR can help the model distinguish the above classes with height differences. LiDAR provides elevation information for distinguishing ground objects with similar spectral reflectance but different heights, and the quality of its data is rarely affected by weather conditions [8], [9], [10]. However, for LiDAR data, as shown in Fig. 1(b), "synthetic grass" and "running track" have similar heights, and it is difficult to distinguish them using LiDAR alone, but their spectral change curves are quite different. So, because of the complementary properties of HSIs and LiDAR, researchers are committed to

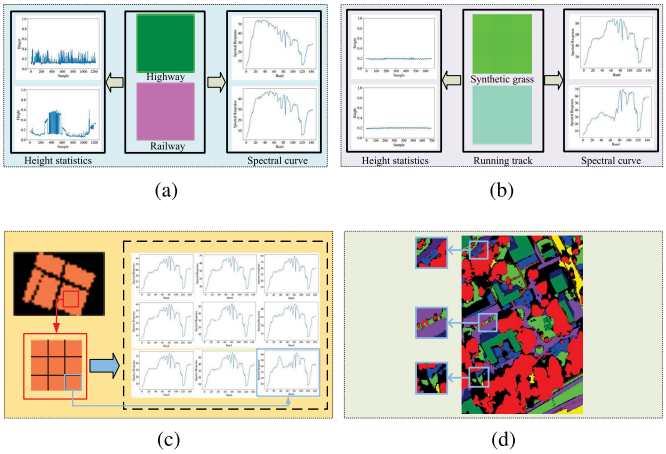


Fig. 1. Challenging scenarios for multimodal data fusion classification. (a) LiDAR is beneficial for distinguishing between ground objects that exhibit similar spectral curves. (b) HSI is beneficial for distinguishing between ground objects that exhibit similar heights. (c) Same class of ground objects has different spectra caused by spectral noise. (d) Spatial distribution of ground objects exhibits complex scale changes.

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The authors are with the Engineering Research Center of Intelligent Control for Underground Space, Ministry of Education, and the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou 221116, China (e-mail: wanghaoyucumt@163.com; xiaominliu@cumt.edu.cn; ts23060226p31@cumt.edu.cn; guoqingwang@cumt.edu.cn; ts23060193p31ty@cumt.edu.cn).

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using HSIs and LiDAR data collaboratively for multimodal remote sensing data classification [11], [12], [13].

Multimodal remote sensing data (HSI and LiDAR) classification methods aim to enhance the model's ability to distinguish ground objects by learning multimodal complementary information. However, the following challenges hinder the performance improvement of existing multimodal remote sensing data classification methods: Difficulty in suppressing spectral noise (Challenge 1), difficulty in fully mining contextual information (Challenge 2), and difficulty in learning the strong adaptive fusion pattern (Challenge 3). Specifically, the resolution of Challenges 1 and 2 can provide pure and precise contextual information features for solving Challenge 3. The solution to Challenge 3 can help the model effectively capture complementary information from multimodal data with significant heterogeneity.

A. Challenge 1: Difficulty in Suppressing Spectral Noise

Due to the influence of weather, light, and other factors, some ground objects in the HSI have severe spectral noise, resulting in different class ground objects having the same spectral or same class ground objects having different spectral, as shown in Fig. 1(c). This phenomenon leads to the model mistakenly establishing statistical correlations between spectral noise and labels, thus causing the model to ignore critical identification information. To address this issue, researchers established feature interactions between HSI and LiDAR based on attention mechanisms, guiding the model to suppress noise information. However, these methods only focus on the utilization of LiDAR elevation information, while ignoring its rich spatial prior information.

Considering because interference of spectral noise, the collected HSI presents an inaccurate spatial topology structure: The spatial continuity of the same class ground objects is poor, and the spatial boundary between different class ground objects is not clear. In contrast, LiDAR can accurately reflect the spatial topology structure of ground objects under different weather and lighting conditions [11]: The spatial continuity of the same class of ground objects is strong, and the spatial boundary between different class ground objects is clear. The spatial topology constraint based on Gaussian mixture (STC-GM) was proposed, using the accurate spatial topology structure of LiDAR as prior information to guide the model suppresses spectral noise that leads to inaccurate spatial topology structure of HSIs.

B. Challenge 2: Difficulty in Fully Mining Contextual Information

In real-world scenarios, the spatial distribution of ground objects often exhibits complex scale changes. As shown in Fig. 1(d), for small-scale ground objects, attention should be paid to their spectral details in the fusion process. In contrast, when dealing with large-scale ground objects, both spectral details and spatial topology structure information should be taken into account. The above phenomenon leads to the difficulty of fully mining the contextual information in multimodal data, which in turn impairs the data fusion effect. Researchers have introduced multiscale techniques to enhance the model's contextual mining ability by integrating information from ground objects at multiple scales. However, existing methods still have certain limitations: 1) ignoring the mutual interference between information of different scale features and 2) same multiscale information fusion parameters are difficult to adapt to the complex scale changes.

The multiscale dynamic graph aggregation module (MDGAM) was proposed to achieve the full mining of two-modal contextual information from two perspectives. On the one hand, each scale feature is regarded as a graph node, and the class identification information that can be shared among multiscale features is captured. Thus, the information interference between multiscale features is mitigated. On the other hand, MDGAM broke through the limitation of fixed fusion parameters and customized multiscale graph aggregation parameters for different scales.

C. Challenge 3: Difficulty in Learning the Strong Adaptive Fusion Pattern

Due to the different imaging mechanisms, there exists modal heterogeneity between the two modalities. This modal heterogeneity leads to the mutual interference of different modal information, which increases the difficulty of strong adaptive fusion pattern learning. Existing methods learn multimodal data fusion patterns by establishing feature interactions between multimodal data. Although these methods have achieved impressive performance, there are still some limitations: 1) it is difficult to suppress intermodal information interference induced by only establishing feature interaction and 2) the learning of multimodal data fusion pattern depends on the observation of limited samples in batch training, so it is difficult to ensure the adaptability of the model.

To address this issue, the multimodal variational autoencoder (M-VAE) is proposed to learn the strong adaptive fusion pattern. The evidence lower bound of the multimodal joint distribution is derived, and the multimodal joint distribution is learned by optimizing this evidence lower bound. Since the multimodal joint distribution directly models the global complementary fusion pattern of multimodal data through the variational inference mechanism [14], [15], the impact of modal heterogeneity on multimodal data fusion can be effectively suppressed.

The main contributions of this article are as follows.

1) M-VAE is proposed, which differs from existing fusion methods that directly establish feature interaction between modalities. M-VAE models multimodal joint distributions for multimodal data fusion, avoiding information interference between modalities caused by modal heterogeneity.
2) MDGAM is proposed, which differs from existing methods that use the same multiscale information fusion parameters for different ground objects. MDGAM dynamically customizes personalized fusion parameters for each ground object, improving the model's adaptability to real-world scenarios with complex scale changes.
3) STC-GM is proposed, which differs from existing methods that often ignore spatial topology information of

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LiDAR. STC-GM utilizes this rich information to constrain the feature learning process, achieving spectral noise suppression.

II. Related Work

A. Spectral Noise Suppression

To suppress the spectral noise in HSIs, a widely used method is the attention mechanism [16]. The attention mechanism suppresses noise by adjusting the model's attention to different attributes of HSI features. Fan et al. [17] designed a composite attention module containing spectral attention and spatial attention, which assigned weights based on pixel-nearest-neighbor relationships and spectral band importance. Wang et al. [18] captured second-order feature interactions between multimodal data to obtain accurate attention weight estimation, thereby obtaining pure multimodal features. Li et al. [19] proposed a center spectrum correction module that achieved noise interference suppression by reducing the importance of different adjacent objects.

However, most of these methods have failed to explore the spatial topology structures of LiDAR. This prevents them from further improving their performance in multimodal remote sensing data classification tasks.

B. Contextual Information Mining

Both HSIs and LiDAR contain rich contextual information that reflects the spatial dependence relationship between ground objects, which directly the performance of the fusion classification model. CNN, which stacks convolutional layers to expand receptive fields, is widely used for spatial information extraction in multimodal fusion classification tasks. Zhang et al. [20] designed a three-branch convolution that learns spectral, spatial, and elevation features, establishing the interaction between different levels of features, the model can improve the utilization ability of multimodal spatial information. Xu et al. [21] used CNNs and LSTM to extract local and global spatial information of HSIs, fully capturing the spatial dependency of ground objects.

The CNN-based fusion classification methods mentioned above are limited by regular convolution kernels, making it difficult to fully capture the spatial dependency relationships between irregular ground objects. Therefore, graph neural networks (GCNs), which are suitable for irregular structural data, have become a noteworthy method. Xiu et al. [22] explored the graph topological structure hidden in the samples. The information transfer mechanism of the GCN is used to capture the irregular spatial neighborhood information. Du et al. [23] constructed a multimodal graph to fuse HSIs and LiDAR spatial information, which ensured that the adjacent ground objects of the same spatial topology region were close to each other in the feature space.

The above methods only utilize spatial information from a single scale, which often fails to fully reflect the complex spatial distribution of ground objects [24]. Multiscale feature fusion technology can integrate information from multiple spatial scales, helping the model fully capture the complex spatial dependencies between ground objects. Ding et al. [10] obtained samples at multiple spatial scales by selecting different numbers of neighborhoods to be classified. And the multiscale spatial information is aggregated by grouping fusion. Li et al. [25] used convolution kernels of different sizes to obtain multiscale spatial information of two modalities, thereby capturing the dependency between multiscale features.

The performance of the above methods is limited by different scale features mutual interference and fixed fusion parameters, and it is difficult to adapt to the scale changes of ground objects.

C. Multimodal Data Fusion

The most widely used data fusion methods for multimodal fusion classification are decision fusion and feature fusion. Decision fusion improves the accuracy of the final decision by integrating the classification decisions of multimodal data. Yan et al. [26] adopted a dual-branch architecture of HSIs and LiDAR, using each branch to independently predict ground object classes. The predictions of the two branches were connected and input into the convolutional layer for fusion predictions. Zhang et al. [27] used a decision fusion strategy based on class label level and posterior probability level to integrate classification decisions from different modalities. Feature fusion aims to obtain multimodal fusion features by integrating multimodal information in feature space. Hu et al. [28] used the cross-attention mechanism to capture modal-sharing information and fused multimodal features. Li et al. [29] first attempted to utilize the representation learning of remote sensing knowledge graphs to generate semantic representations of remote sensing scene categories. By conducting robust cross-modal alignment between visual features and semantic representations, they effectively mitigated the interference between modal information that could be induced by modal heterogeneity. Hang et al. [30] used the cross-reconstruction between HSIs and LiDAR features to explore the intrinsic semantic and structural information of training samples. Then, the modal sharing feature representation is obtained. Dong et al. [31] proposed a multibranch feature fusion network to obtain multimodal fusion features by simple cat HSI and LiDAR branching features. Li et al. [32] proposed the distribution balancing subconstraint and the variance maximization subconstraint, which promote cross-modal knowledge sharing reducing the multimodal class structure differences between the visual image space and the semantic space. Yao et al. [33] proposed a deep-learning-based multimodal remote sensing data classification method, which contains the cross-modal multitask learning mechanism and the cross-modal attention mechanism. This method not only solved the problem of modal incompleteness in multimodal learning but also realized efficient information exchange between different modalities. Hong et al. [34] integrated the rich spatial information of multimodal remote sensing data by different-resolution feature fusion and extended the common two-modal fusion to three-modal scenarios. In addition, they presented a complex multimodal remote sensing dataset with challenges. Song et al. [35] designed a novel hashing-based deep metric learning method, which used similarity measures

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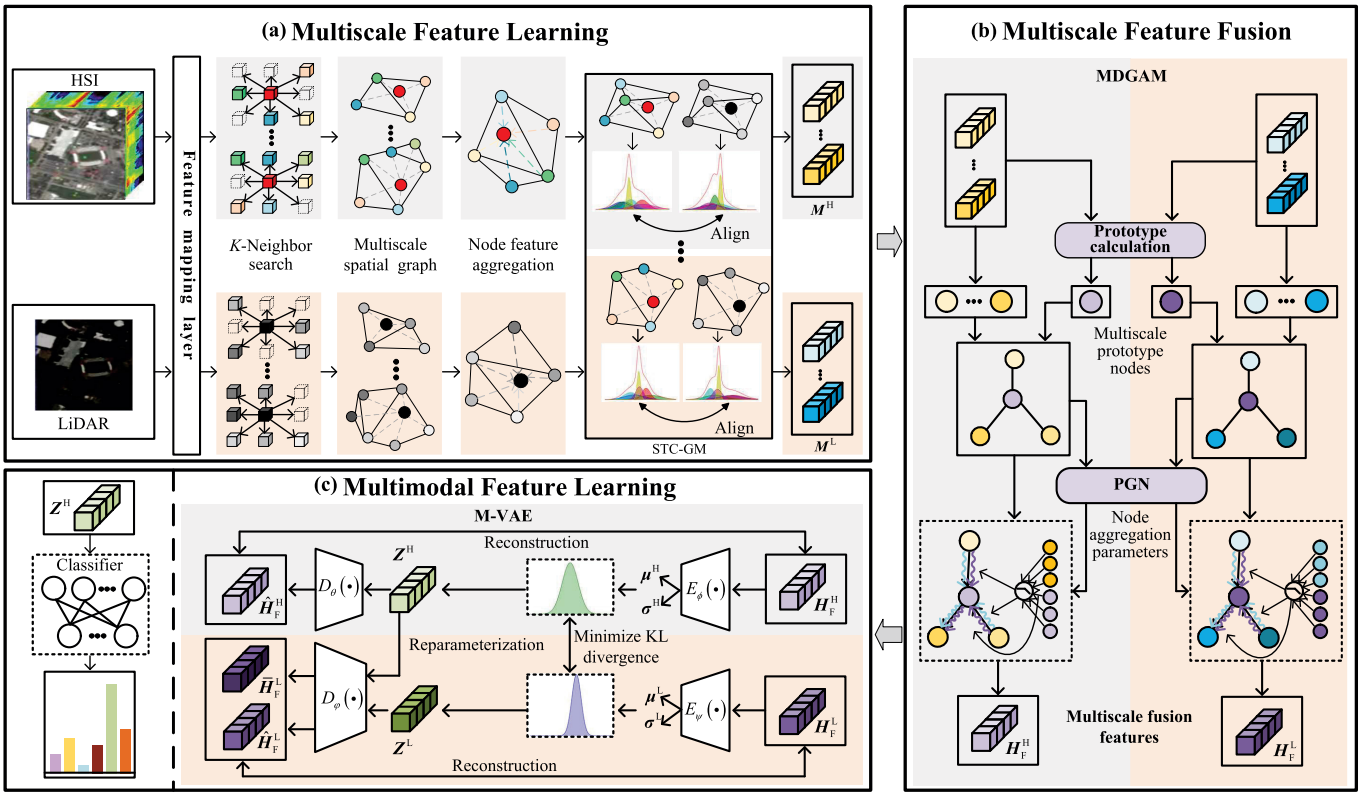


Fig. 2. Framework of GM-VDFN. (a) Two-modal multiscale spatial graphs were constructed to input GCN for node feature learning, and pure two-modal multiscale features were obtained with the constraint of STC-GM. (b) MDGAM is used to fuse the pure two-modal multiscale features separately, obtaining the multiscale fusion features of the two modalities. (c) Multiscale fusion features of the two modalities were input into M-VAE to model multimodal joint distributions, and $\boldsymbol{Z}^{H}$ containing multimodal joint distribution information was input into the classifier for class prediction.

based on Hashing to excavate correlation information. This method made full use of the complementary and related information of HSI and LiDAR data from the perspective of metric learning. Song et al. [36] proposed a deep feature fusion network, which used residual learning to solve the problem of gradient vanishing during the multimodal feature learning process, and integrated features of different hierarchical layers to achieve deep multimodal data fusion.

However, most of the above methods ignore the information interference between the modalities and fail to model the joint distribution between HSI and LiDAR. It makes it difficult to ensure that the key complementary information of both modalities is retained during multimodal data fusion.

III. GM-VDFN for Multimodal Remote Sensing Data Classification

A. Framework of GM-VDFN

The training process of the Gaussian mixture variational dynamic fusion network (GM-VDFN) is shown in Fig. 2, which can be divided into the following steps.

1) Multiscale Feature Learning: First, the K-nearest neighbor search is used to divide HSI and LiDAR into K-nearest neighbor ground objects clusters. Second, HSI and LiDAR single-scale spatial graphs are constructed based on the K-nearest ground objects cluster. Third, multiscale spatial graphs of HSI and LiDAR are obtained by adjusting the number of K neighbors and repeating the above process. Fourth, each scale spatial graph of HSI and LiDAR is input into the GCN for node feature aggregation, and the node features of each node are updated. Fifth, calculate the spatial topology matrix of two modalities based on node features and model the spatial topology matrix as a Gaussian mixture distribution. Finally, the STC-GM is used to align the spatial distributions of two domains and then force the nodes in the same topological region of two modalities to have similar features, thereby the pure two-modal multiscale feature set is obtained.

2) Multiscale Feature Fusion: First, the pure two-modal multiscale feature set is input into the MDGAM. MDGAM treats the features of each scale as graph nodes and obtains two-modal multiscale prototype nodes through prototype calculation. Second, the connection relationship between multiscale prototype nodes and each scale node is established, constructing a two-modal multiscale prototype graph. Finally, the parameter-generated network (PGN) generates node aggregation parameters dynamically based on the edge features between the multiscale prototype node and each scale node. The information transfer mechanism is used to aggregate the multiscale node information and the multiscale fusion features of two modalities are obtained.

3) Multimodal Feature Learning: First, the multiscale fusion features of the two modalities are input into the M-VAE. By optimizing the evidence lower bound of the M-VAE, the joint distribution of the two modalities is modeled, and the multimodal features are obtained. Then, the multimodal

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features are entered into the classifier to complete the remote sensing data classification task.

B. Multiscale Spatial Graph Construction

To achieve computational acceleration and batch learning similar to miniGCN [37], we first cut the remote sensing data into several patches centered on the target pixels to be classified and then constructed the spatial graphs of each patch. In both HSI and LiDAR, spatial adjacent objects with high spatial consistency and feature similarity have a high probability of belonging to the same class [38]. Based on this spatial prior, the K-nearest neighbor composition is used [39] to segment HSI and LiDAR into clusters, to make full use of the spatial contextual information around the object (to be classified). When $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ are K neighbors, an edge connection is created between nodes can be expressed as

$$\boldsymbol{A}_{i,j} = \begin{cases} D(\boldsymbol{x}_i, \boldsymbol{x}_j), & \text{if } \boldsymbol{x}_i \in N_{\boldsymbol{x}_j} \text{ and } \boldsymbol{x}_j \in N_{\boldsymbol{x}_i} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $N_{\boldsymbol{x}}$ represents the set of K neighbors of $\boldsymbol{x}$, and $D(\cdot)$ represents the measure of similarity between ground objects, which can be expressed as

$$D(\boldsymbol{x}_i, \boldsymbol{x}_j) = \frac{\sqrt{\left(\text{cord}(\boldsymbol{x}_i) - \text{cord}(\boldsymbol{x}_j)\right)^T \left(\text{cord}(\boldsymbol{x}_i) - \text{cord}(\boldsymbol{x}_j)\right)}}{\cos\left(\frac{\boldsymbol{x}_i \cdot \boldsymbol{x}_j}{\|\boldsymbol{x}_i\|_2 \|\boldsymbol{x}_j\|_2}\right)} \tag{2}$$

where $\text{cord}(\boldsymbol{x}_i)$ and $\text{cord}(\boldsymbol{x}_j)$ represent the spatial coordinates of $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, respectively. The above measure considers the spectral similarity and spatial distance between ground objects, which can comprehensively reflect the dependencies between ground objects.

Then, according to the adjacency matrix $\boldsymbol{A}$, the edge set E of the single-scale spatial graph is obtained. Each ground object is treated as a graph node to obtain the node set V, and the single-scale spatial graph $G = (V, E)$ is obtained.

Finally, by setting different K-nearest neighbor numbers during the construction of HSI and LiDAR spatial graphs, a two-modal multiscale spatial graph set can be obtained

$$\begin{cases} G^H(V^H, E^H) = \left\{G^H(V_0^H, E_0^H), \ldots, G^H(V_S^H, E_S^H)\right\} \\ G^L(V^L, E^L) = \left\{G^L(V_0^L, E_0^L), \ldots, G^L(V_S^L, E_S^L)\right\} \end{cases} \tag{3}$$

where $G^H(V^H, E^H)$ is the multiscale spatial graph set of HSIs, $G^L(V^L, E^L)$ is the multiscale spatial graph set of LiDAR, and S is the number of scales.

C. Feature Extraction Module Based on GCN

First, given a graph $G = (V, E)$ and its corresponding adjacency matrix $\boldsymbol{A}$, the corresponding Laplacian matrix $\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{A}$ can be obtained, where $\boldsymbol{D}$ is the degree matrix of $\boldsymbol{A}$ and can be expressed as $\boldsymbol{D}_{i,i} = \sum_j \boldsymbol{A}_{i,j}$.

Then, the normalized Laplacian matrix is obtained

$$\boldsymbol{L}_n = \boldsymbol{I} - \boldsymbol{D}^{-\frac{1}{2}} \boldsymbol{A} \boldsymbol{D}^{-\frac{1}{2}} \tag{4}$$

where $\boldsymbol{I}$ is the identity matrix. After spectral decomposition of the normalized Laplace matrix, $\boldsymbol{L}_n = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{U}^T$ can be obtained, where $\boldsymbol{U}$ is the set of eigenvectors in $\boldsymbol{L}_n$, and $\boldsymbol{\Sigma}$ is the diagonal matrix of $\boldsymbol{L}_n$. The convolution operation on the graph can be expressed as

$$\boldsymbol{v} * \boldsymbol{f}_\theta = \boldsymbol{U} \boldsymbol{f}_\theta(\boldsymbol{\Sigma}) \boldsymbol{U}^T \boldsymbol{v} \tag{5}$$

where $\boldsymbol{v}$ represents the node feature, $\boldsymbol{f}_\theta$ can be regarded as a function of normalized Laplace matrix eigenvalues, $\boldsymbol{\Sigma}$, and θ is a learnable parameter. Since the calculation of the above equation is difficult, it is approximated using the first-order expansion of the Chebyshev polynomial [40]

$$\boldsymbol{v} * \boldsymbol{f}_\theta \approx \theta\left(\boldsymbol{I} + \boldsymbol{D}^{-\frac{1}{2}} \boldsymbol{A} \boldsymbol{D}^{-\frac{1}{2}}\right). \tag{6}$$

Finally, the propagation rules for the GCN are available

$$\boldsymbol{H}^{(l+1)} = \boldsymbol{\sigma}\left(\tilde{\boldsymbol{D}}^{-\frac{1}{2}} \tilde{\boldsymbol{A}} \tilde{\boldsymbol{D}}^{-\frac{1}{2}} \boldsymbol{H}^{(l)} \boldsymbol{W}^{(l+1)} + \boldsymbol{b}^{(l+1)}\right) \tag{7}$$

where $\tilde{\boldsymbol{A}} = \boldsymbol{A} + \boldsymbol{I}$, $\tilde{\boldsymbol{D}}$ is the degree matrix of $\tilde{\boldsymbol{A}}$, $\boldsymbol{\sigma}$ is the activation function, $\boldsymbol{H}^{(l)}$ is the output of the l-layer graph convolution, and $\boldsymbol{W}^{(l+1)}$ and $\boldsymbol{b}^{(l+1)}$ are the learnable weight matrix and bias term in the $(l+1)$-layer graph convolution, respectively.

The input of the feature extraction module is a multiscale spatial graph set of two modalities, which outputs multiscale feature sets $\boldsymbol{M}^H$ and $\boldsymbol{M}^L$ of HSIs and LiDAR

$$\begin{cases} \boldsymbol{M}^H = \left[\boldsymbol{H}_1^H, \ldots, \boldsymbol{H}_S^H\right] \\ \boldsymbol{M}^L = \left[\boldsymbol{H}_1^L, \ldots, \boldsymbol{H}_S^L\right]. \end{cases} \tag{8}$$

D. Spectral Noise Suppression Based on STC-GM

HSIs can provide rich spectral information and help distinguish ground objects with different spectral properties. However, it is susceptible to spectral noise, making it difficult for a model to distinguish different ground objects. To solve this problem, STC-GM is proposed. The spatial prior that ground objects in the same topology region with a high probability of belonging to the same class, and the precise spatial topology structure of LiDAR are used to constrain the features of the ground objects, realizing the spectral noise suppression. The spatial distribution of HSI and LiDAR is modeled as a Gaussian mixture distribution. By explicitly constraining ground objects in the same topological region to have consistent features, spectral noise is suppressed.

As shown in Fig. 3, first, to describe the spatial topology structures of the two modalities (HSI and LiDAR), the similarity between each pixel feature and the central pixel feature in the image patch is calculated to obtain the spatial topology matrices and obtain the spatial topology matrices $\boldsymbol{C}^H$ and $\boldsymbol{C}^L$ of HSI and LiDAR

$$\boldsymbol{C}_i = \frac{\sum_{e=1}^{E} \boldsymbol{v}_i^e \times \boldsymbol{v}_c^e}{\sqrt{\sum_{e=1}^{E} \boldsymbol{v}_i^e} \times \sqrt{\sum_{e=1}^{E} \boldsymbol{v}_c^e}} \tag{9}$$

where $\boldsymbol{v}_i^e$ and $\boldsymbol{v}_c^e$ represent the e-dimensional features of pixel $\boldsymbol{v}_i$ and central pixel $\boldsymbol{v}_c$ extracted by feature extraction module respectively, and E is the node feature dimension.

Second, based on the spatial topology matrices, the Gaussian mixture model is used to model the spatial distributions (green) of the two modalities. The spatial distribution includes

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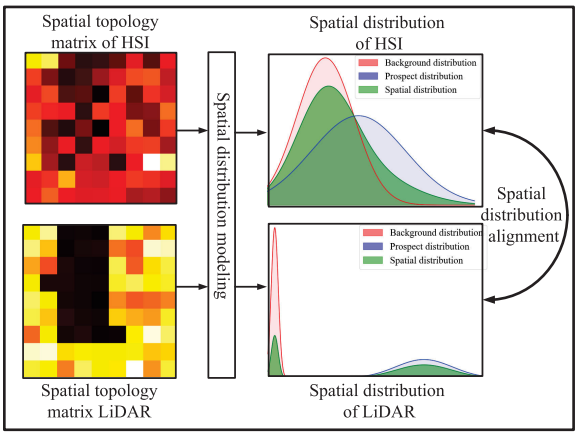


Fig. 3. Flowchart of the proposed STC-GM.

the prospect distribution (blue) and background distribution (red). The former represents the spatial distribution of pixels of the same class as the central pixel, and the latter represents the spatial distribution of pixels of a different class from the central pixel. Due to the influence of spectral noise, the spatial continuity of the prospect in HSIs is poor, and the spatial boundaries between the prospect and background are difficult to distinguish. The former leads to a high variance in the prospect distribution, while the latter results in an unimodal spatial distribution due to the mixing of prospect distribution and background distribution. In contrast, the spatial continuity of the prospect in LiDAR pixels is high, and the spatial boundary between the prospect and background is clear. At this point, the prospect distribution shows low variance, while the spatial distribution shows a bimodal pattern due to the separation of the prospect and background distributions.

We randomly initialize a set of Gaussian distribution mean matrix, variance matrix, and their mixture weight of prospect distribution and background distribution. Then, the expectation maximization is used to optimize the parameters of the Gaussian mixture distribution.

In the Expectation step, parameters of the Gaussian distribution are fixed, and the posterior probability that each data point in the spatial topology matrix comes from each Gaussian distribution is calculated. For data point C_b and Gaussian distribution $N(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$, the posterior probability is calculated as

$$\boldsymbol{\gamma}_{bj} = \frac{\boldsymbol{\Psi}_j N\left(\boldsymbol{C}_b|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j\right)}{\sum_{m=1}^{M} \boldsymbol{\Psi}_m N\left(\boldsymbol{C}_b|\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m\right)} \tag{10}$$

where $\boldsymbol{\Psi}_j$ is the mixed weight of the Gaussian distribution $N(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$, and $N(\boldsymbol{C_b}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$ is the probability density of $N(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$ at $\boldsymbol{C}_b$.

In the Maximization step, with a fixed posterior probability, update the parameters of each Gaussian distribution as follows:

$$\begin{cases} \boldsymbol{\mu}_j = \dfrac{\sum_{b=1}^{B} \boldsymbol{\gamma}_{bj} \boldsymbol{C}_b}{\sum_{b=1}^{B} \boldsymbol{\gamma}_{bj}} \\ \boldsymbol{\Sigma}_j = \dfrac{\sum_{b=1}^{B} \boldsymbol{\gamma}_{bj} \left(\boldsymbol{C}_b - \boldsymbol{\mu}_j^{\text{new}}\right)\left(\boldsymbol{C}_b - \boldsymbol{\mu}_j^{\text{new}}\right)^T}{\sum_{b=1}^{B} \boldsymbol{\gamma}_{bj}} \\ \boldsymbol{\Psi}_j = \dfrac{1}{B} \sum_{b=1}^{B} \boldsymbol{\gamma}_{bj} \end{cases} \tag{11}$$

where B is the number of data points in the spatial topology matrix.

The Expectation step and Maximization step are repeated until the method converges, and then the Gaussian mixture distribution of the spatial topology matrix is obtained

$$p(\boldsymbol{C}_b) = \sum_{m=1}^{M} \boldsymbol{\Psi}_m \frac{1}{(2\pi)^{\frac{1}{2}} |\boldsymbol{\Sigma}_m|^{\frac{1}{2}}} e^{-\frac{1}{2}(\boldsymbol{C}_b - \boldsymbol{\mu}_m)^T \boldsymbol{\Sigma}_m^{-1} (\boldsymbol{C}_b - \boldsymbol{\mu}_m)}. \tag{12}$$

Finally, using the spatial distribution of LiDAR as prior information, the variance of the HSI prospect distribution is reduced through spatial distribution alignment, while promoting the separation of its prospect and background distributions. Correspondingly, STC-GM constrains the output features of the feature extraction module during model training, guiding the model to suppress spectral noise through backpropagation. This operation enhances the spatial continuity of the same class of ground objects and enhances the distinguishability of spatial boundaries between different class ground objects in HSIs. Specifically, spatial distribution alignment is achieved by minimizing the KL divergence between $p^H = \sum_{m=1}^{M} \boldsymbol{\Psi}_m^H N(\boldsymbol{C}^H|\boldsymbol{\mu}_m^H, \boldsymbol{\Sigma}_m^H)$ and $p^L = \sum_{m=1}^{M} \boldsymbol{\Psi}_m^L N(\boldsymbol{C}^L|\boldsymbol{\mu}_m^L, \boldsymbol{\Sigma}_m^L)$. We use Monte Carlo integrals to approximate the KL divergence between p^H and p^L, and the spatial topology constraint L_s can be expressed as

$$L_s = D_{\text{KL}}\left(p^L || p^H\right) \approx \frac{1}{B} \sum_{b=1}^{B} p^L\left(\boldsymbol{C}_b^L\right) \log \frac{p^L\left(\boldsymbol{C}_b^L\right)}{p^H\left(\boldsymbol{C}_b^H\right)}. \tag{13}$$

E. Contextual Information Fully Mining Based on MDGAM

MDGAM aggregates multiscale features to capture multilevel spatial structure and contextual information from rough to fine, which realizes accurate descriptions of ground objects at different scales.

First, each scale feature in the multiscale feature set $\boldsymbol{M} = \{\boldsymbol{H}_1, \ldots, \boldsymbol{H}_K\}$ is regarded as graph nodes. The edge connections are established between these nodes and multiscale prototype node $\boldsymbol{P} = (1/K) \sum_{k=1}^{K} \boldsymbol{H}_k$, constructing a two-modal multiscale prototype graph.

Second, the attention coefficient between $\boldsymbol{P}$ and each scale node is calculated, and the attention coefficient $\boldsymbol{\alpha}_k$ of the scale k node can be expressed as

$$\boldsymbol{\alpha}_k = \frac{\exp(\text{LeakyReLU}(\boldsymbol{e}_k))}{\sum_{i=1}^{K} \exp(\text{LeakyReLU}(\boldsymbol{e}_i))}, \quad \boldsymbol{e}_k = \boldsymbol{W}^A([\boldsymbol{W}\boldsymbol{H}_k || \boldsymbol{W}\boldsymbol{P}]) \tag{14}$$

where $\boldsymbol{W}^A$ is the attention matrix and $\boldsymbol{W}$ is the node mapping matrix.

Third, based on the edge feature $\boldsymbol{e}_k$, the node aggregation parameters are dynamically generated to aggregate the multiscale features, obtaining the multiscale fusion feature $\boldsymbol{H}_F$

$$\boldsymbol{H}_F = \sigma\left(\sum_{k=1}^{K} \boldsymbol{\alpha}_k \widehat{\boldsymbol{W}}_k \boldsymbol{H}_k\right), \quad \widehat{\boldsymbol{W}}_k = \text{PGN}(\boldsymbol{e}_k) \tag{15}$$

where $\sigma(\cdot)$ is the activation function and $\text{PGN}(\cdot)$ is the parameter generation network, which is composed of multilayer perception. The input of PGN is an edge feature that can reflect the dependency between multiscale prototype nodes

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and multiscale nodes. The output is the aggregation parameter corresponding to the multiscale node.

The input of MDGAM is a multiscale feature set of two modalities, and the output is the multiscale fusion feature $\boldsymbol{H}_F^H$ of HSI and the multiscale fusion feature $\boldsymbol{H}_F^L$ of LiDAR.

F. Multimodal Feature Fusion Based on M-VAE

To overcome the heterogeneity between HSI and LiDAR and obtain high-quality multimodal features, M-VAE is proposed. The multimodal data fusion is transformed into a multimodal joint distribution modeling. Thus, the powerful distribution modeling ability of VAE is utilized to model the multimodal joint distribution of HSIs and LiDAR. Specifically, M-VAE derives the evidence lower bound for the multimodal joint distribution of HSIs and LiDAR and learns the multimodal joint distribution by optimizing the evidence lower bound. Furthermore, high-quality fusion features that can capture multimodal joint distribution are obtained in the hidden space.

The evidence lower bound of multimodal joint distribution can be expressed as

$$\begin{aligned}&\log p_{\theta,\varphi}\left(\boldsymbol{H}_F^L,\boldsymbol{H}_F^H\right)\\&\quad=\log\int_{\boldsymbol{Z}}p_{\theta,\varphi}\left(\boldsymbol{H}_F^L,\boldsymbol{H}_F^H,\boldsymbol{Z}\right)d\boldsymbol{Z}\\&\quad\geq\int_{\boldsymbol{Z}}\frac{p_{\theta,\psi}\left(\boldsymbol{H}_F^L,\boldsymbol{H}_F^H,\boldsymbol{Z}\right)}{q_{\phi,\varphi}\left(\boldsymbol{Z}|\boldsymbol{H}_F^L,\boldsymbol{H}_F^H\right)}q_{\phi,\varphi}\left(\boldsymbol{Z}|\boldsymbol{H}_F^L,\boldsymbol{H}_F^H\right)d\boldsymbol{Z}.\end{aligned}\tag{16}$$

Use conditional independence $\boldsymbol{H}_F^L\perp\boldsymbol{H}_F^H|\boldsymbol{Z}$ [41] and Bayer rule to model $q_{\phi,\varphi}(\boldsymbol{Z}|\boldsymbol{H}_F^L,\boldsymbol{H}_F^H)$

$$\begin{cases}q_{\phi,\varphi}\left(\boldsymbol{Z}|\boldsymbol{H}_F^L,\boldsymbol{H}_F^H\right)=\dfrac{q_\phi\left(\boldsymbol{Z}|\boldsymbol{H}_F^H\right)q_\varphi\left(\boldsymbol{H}_F^L|\boldsymbol{Z}\right)}{q_{\phi,\varphi}\left(\boldsymbol{H}_F^L|\boldsymbol{H}_F^H\right)}\\q_{\phi,\varphi}\left(\boldsymbol{H}_F^L|\boldsymbol{H}_F^H\right)=\displaystyle\int q_\phi\left(\boldsymbol{Z}|\boldsymbol{H}_F^H\right)q_\varphi\left(\boldsymbol{H}_F^L|\boldsymbol{Z}\right)d\boldsymbol{Z}.\end{cases}\tag{17}$$

Thus, the new evidence lower bound can be expressed as

$$\begin{aligned}&\log p_{\theta,\psi}\left(\boldsymbol{H}_F^L,\boldsymbol{H}_F^H\right)\geq L_{F\{\Theta,\Psi\}}\left(\boldsymbol{H}_F^L,\boldsymbol{H}_F^H\right)\\&=\int_{\boldsymbol{Z}}\frac{p_{\theta,\psi}\left(\boldsymbol{H}_F^H,\boldsymbol{H}_F^L,\boldsymbol{Z}\right)q_{\phi,\varphi}\left(\boldsymbol{H}_F^L|\boldsymbol{H}_F^H\right)}{q_\phi\left(\boldsymbol{Z}|\boldsymbol{H}_F^H\right)q_\varphi\left(\boldsymbol{H}_F^L|\boldsymbol{Z}\right)}\frac{q_\phi\left(\boldsymbol{Z}|\boldsymbol{H}_F^H\right)q_\varphi\left(\boldsymbol{H}_F^L|\boldsymbol{Z}\right)}{q_{\phi,\varphi}\left(\boldsymbol{H}_F^L|\boldsymbol{H}_F^H\right)}d\boldsymbol{Z}\\&=\mathbb{E}_{q_\phi\left(\boldsymbol{Z}|\boldsymbol{H}_F^H\right)}\left[\frac{q_\varphi\left(\boldsymbol{H}_F^L|\boldsymbol{Z}\right)}{q_{\phi,\varphi}\left(\boldsymbol{H}_F^L|\boldsymbol{H}_F^H\right)}\log\frac{p_\theta\left(\boldsymbol{H}_F^H|\boldsymbol{Z}\right)p_\psi\left(\boldsymbol{Z}|\boldsymbol{H}_F^L\right)}{q_\phi\left(\boldsymbol{Z}|\boldsymbol{H}_F^H\right)q_\varphi\left(\boldsymbol{H}_F^L|\boldsymbol{Z}\right)}\right]\\&\quad+\log q_{\phi,\varphi}\left(\boldsymbol{H}_F^L|\boldsymbol{H}_F^H\right)+\log p\left(\boldsymbol{H}_F^L\right)\end{aligned}\tag{18}$$

where $\Theta=\{\theta,\psi\}$ and $\Psi=\{\phi,\varphi\}$. ϕ and θ are the encoder and decoder parameters for $\boldsymbol{H}_F^H$, ψ and φ are the encoder and decoder parameters for $\boldsymbol{H}_F^L$. $\mathbb{E}_{q_{\phi_{(\boldsymbol{Z}|\boldsymbol{H}_F^L)}}}[(q_\varphi(\boldsymbol{H}_F^L|\boldsymbol{Z}))/(q_{\phi,\varphi}(\boldsymbol{H}_F^L|\boldsymbol{H}_F^H))\log(p_\theta(\boldsymbol{H}_F^H|\boldsymbol{Z})p_\psi(\boldsymbol{Z}|\boldsymbol{H}_F^L))/(q_\phi(\boldsymbol{Z}|\boldsymbol{H}_F^H)q_\varphi(\boldsymbol{H}_F^L|\boldsymbol{Z}))]$ is the weighted likelihood error of two-modal reconstruction. On the one hand, the encoder and the decoder are trained by two-modal feature reconstruction. On the other hand, the KL divergence of the two modal posterior distributions $p_\psi(\boldsymbol{Z}|\boldsymbol{H}_F^L)$ and $p_\psi(\boldsymbol{Z}|\boldsymbol{H}_F^L)$ is minimized, so that the model learns the modal shareable features. $\log q_{\phi,\varphi}(\boldsymbol{H}_F^L|\boldsymbol{H}_F^H)$ uses $\boldsymbol{H}_F^H$ to pseudo-reconstruct $\boldsymbol{H}_F^L$, which establishes information interaction between two modalities through modal transformation. $\log p(\boldsymbol{H}_F^L)$ is a constant and can be ignored during optimization. Based on this evidence lower bound, we construct the M-VAE that contains the two-modal encoder, two-modal decoder, and sample weighted module.

First, $\boldsymbol{H}_F^H$ and $\boldsymbol{H}_F^L$ are entered into encoders E_ϕ and E_ψ, respectively, obtaining the mean and variance of the posterior distribution

$$\begin{cases}\boldsymbol{\mu}^H,\quad\boldsymbol{\sigma}^H=E_\phi\left(\boldsymbol{H}_F^H\right)\\\boldsymbol{\mu}^L,\quad\boldsymbol{\sigma}^L=E_\psi\left(\boldsymbol{H}_F^L\right).\end{cases}\tag{19}$$

Second, the reparameterization [42] technique is used to obtain the two-modal latent variables $\boldsymbol{Z}^H$ and $\boldsymbol{Z}^L$

$$\begin{cases}\boldsymbol{Z}^H=\boldsymbol{\mu}^H+\boldsymbol{\sigma}^H*\boldsymbol{\varepsilon}^H\\\boldsymbol{Z}^L=\boldsymbol{\mu}^L+\boldsymbol{\sigma}^L*\boldsymbol{\varepsilon}^L\end{cases}\tag{20}$$

where $\boldsymbol{\varepsilon}^H$ and $\boldsymbol{\varepsilon}^L$ are sampled from the standard multivariate Gaussian distribution.

Third, the two-modal latent variables are input into the decoders $\boldsymbol{D}_\theta$ and $\boldsymbol{D}_\varphi$, thus $\boldsymbol{H}_F^H$ and $\boldsymbol{H}_F^L$ are reconstructed

$$\begin{cases}\hat{\boldsymbol{H}}_F^H=\boldsymbol{D}_\theta\left(\boldsymbol{Z}^H\right)\\\hat{\boldsymbol{H}}_F^L=\boldsymbol{D}_\varphi\left(\boldsymbol{Z}^L\right).\end{cases}\tag{21}$$

In addition, to capture the complementary properties of the two modalities, the latent variable corresponding to $\boldsymbol{H}_F^H$ is input into decoder $\boldsymbol{D}_\varphi$ for reconstructing $\boldsymbol{H}_F^L$. The pseudo-reconstruction feature $\bar{\boldsymbol{H}}_F^L$ is obtained

$$\bar{\boldsymbol{H}}_F^L=\boldsymbol{D}_\varphi\left(\boldsymbol{Z}^H\right).\tag{22}$$

Fourth, calculate the difference between pseudo-reconstructed feature $\bar{\boldsymbol{H}}_F^L$ and reconstruct feature $\hat{\boldsymbol{H}}_F^L$ for weighting the sample. Thus, the posterior distribution $q_\phi(\boldsymbol{Z}^H|\boldsymbol{H}_F^H)$ can approximate the joint distribution of the two modalities, and the sample weight $\boldsymbol{w}$ can be expressed as

$$\boldsymbol{w}=\exp\left(\log q_\varphi\left(\boldsymbol{H}_F^L|\boldsymbol{Z}^L\right)-\log q_{\phi,\varphi}\left(\boldsymbol{H}_F^L|\boldsymbol{H}_F^H\right)\right).\tag{23}$$

Fifth, the loss of M-VAE can be obtained

$$\begin{aligned}L_F=\boldsymbol{w}\Bigg(&\left\|\hat{\boldsymbol{H}}_F^H-\boldsymbol{H}_F^H\right\|_2^2+\left\|\hat{\boldsymbol{H}}_F^L-\boldsymbol{H}_F^L\right\|_2^2\\&-\mathrm{KL}\left(p_\psi\left(\boldsymbol{Z}^L|\boldsymbol{H}_F^L\right)||q_\phi\left(\boldsymbol{Z}^H|\boldsymbol{H}_F^H\right)\right)\Bigg)+\left\|\bar{\boldsymbol{H}}_F^L-\boldsymbol{H}_F^L\right\|_2^2.\end{aligned}\tag{24}$$

Finally, $\boldsymbol{Z}^H$ containing the multimodal joint distribution information is input to the classifier, obtaining the predicted class of the sample, and the classification loss L_c is calculated

$$L_c=\xi\left(\hat{\boldsymbol{Y}},\boldsymbol{Y}\right),\quad\hat{\boldsymbol{Y}}=\mathrm{Cls}\left(\boldsymbol{Z}^H\right)\tag{25}$$

where $\xi(\cdot)$ represents cross-entropy and $\hat{\boldsymbol{Y}}$ and $\boldsymbol{Y}$ represent predictive labels and true labels, respectively. $\mathrm{Cls}(\cdot)$ represents a classifier network.

G. Total Loss

The total loss of GM-VDFN can be expressed as

$$L=L_c+\lambda_1L_s+\lambda_2L_F\tag{26}$$

where λ_1 and λ_2 are the balance coefficients that control the loss weights.

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TABLE I
NUMBER OF TRAINING AND TEST SETS FOR EACH DATASET

Houston 2013				Houston 2018			
	Class	Sample number			Class	Sample number	
No	Name	Training	Test	No	Name	Training	Test
1	Healthy grass	20	1231	1	Health grass	200	39996
2	Stressed grass	20	1234	2	Stress grass	200	130808
3	Synthetic grass	20	677	3	Artificial turf	200	3536
4	Trees	20	1224	4	Evergreen trees	200	55122
5	Soil	20	1222	5	Deciduous trees	200	20972
6	Water	20	305	6	Bare earth	200	18864
7	Residential	20	1248	7	Water	200	1864
8	Commercial	20	1224	8	Residential buildings	200	16699
9	Road	20	1232	9	Non-residential buildings	200	895569
10	Highway	20	1207	10	Roads	200	184083
11	Railway	20	1215	11	Sidewalks	200	136835
12	Parking lot 1	20	1213	12	Crosswalks	200	6859
13	Parking lot 2	20	449	13	Major thoroughfares	200	186238
14	Tennis court	20	408	14	Highways	200	40238
15	Running track	20	640	15	Railways	200	28548
				16	Paved parking lots	200	46732
				17	Unpaved parking lots	200	637
				18	Cars	200	27089
				19	Trains	200	22279
				20	Stadium seats	200	28096
	Total	300	14729		Total	4000	2033660
	Total (Training + Test)	15029			Total (Training + Test)	2037660	
Trento				MUUFL			
	Class	Sample number			Class	Sample number	
No	Name	Training	Test	No	Name	Training	Test
1	Apple trees	20	4014	1	Trees	20	23226
2	Buildings	20	2883	2	Mostly grass	20	4250
3	Ground	20	459	3	Mixed ground surface	20	6862
4	Woods	20	9103	4	Dirt and sand	20	1806
5	Vineyard	20	10481	5	Road	20	6667
6	Roads	20	3154	6	Water	20	446
				7	Building shadow	20	2213
				8	Building	20	6220
				9	Sidewalk	20	1365
				10	Yellow curb	20	163
				11	Cloth panels	20	249
	Total	120	30094		Total	220	53467
	Total (Training + Test)	30214			Total (Training + Test)	53687	

IV. EXPERIMENT

A. HSI Dataset

Four HSI and LiDAR fusion datasets are selected to verify the effectiveness of the proposed method. The number of training and test set samples in each dataset is shown in Table I.

1) *Houston 2013 Dataset:* The dataset was collected on the campus of the University of Houston in Texas and nearby areas in 2013, which includes a total of 15 029 ground-truth samples, covering 15 ground object classes. The size of the dataset is 349 × 1905 pixels and has a spatial resolution of 2.5 m. HSI consists of 144 different spectral bands with wavelengths ranging from 0.38 to 1.05 μm. The LiDAR data and HSI have the same spatial resolution, presenting researchers with elevation information on the terrain.
2) *Trento Dataset:* The dataset was collected in a rural area south of Trento, Italy. The dataset has a total of 30 214 ground-truth samples, covering six ground object classes. The data size of the dataset is 166 × 600 pixels and has a spatial resolution of 1 m. HSI consists of 63 spectral bands with wavelengths ranging from 0.40 to 0.98 μm. The LiDAR data and HSI have the same spatial resolution, presenting researchers with elevation information on the terrain.
3) *MUUFL Dataset:* The dataset was collected at the University of Southern Mississippi in Long Beach, Mississippi. The dataset has a total of 53 687 ground-truth samples, covering 11 ground object classes. The size of this dataset is 325 × 220 pixels. The spatial resolution of HSI is 0.54 × 1.0 m, while the LiDAR data and HSI have the same spatial resolution, presenting researchers with elevation information of the terrain.
4) The Houston 2018 dataset was collected on the campus of the University of Houston and neighboring urban areas. It comprises 20 classes of ground objects with spatial dimensions of 1202 × 4768, encompassing a total of 2 037 660 labeled samples. This dataset includes HSI and LiDAR data spatially aligned. The HSI consists of 48 spectral bands within the wavelength range of 380–1050 nm. The LiDAR provides precise elevation data reflecting the characteristics of the terrain.

B. Experimental Setting

To evaluate the effectiveness of the proposed method, we compare it with multiple methods, including the CNN [43],

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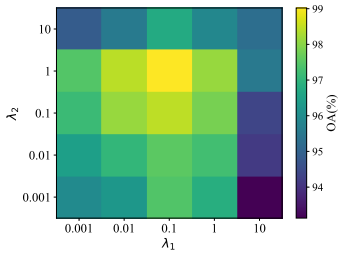


Fig. 4. Variation of OA with different λ_1 and λ_2.

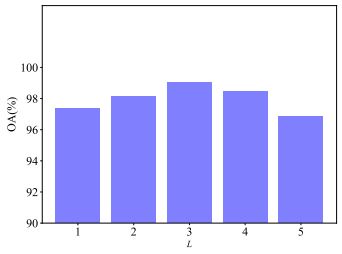


Fig. 5. Variation of OA with different L.

hierarchical random walk network (HRWN) [44], network for endmember extraction and hyperspectral unmixing (EndNet) [45], consistent contrastive representation network for multiview clustering (CCR-Net) [46], and the fractional Gabor convolutional network (FGCN) [47]. All experiments were conducted on a server equipped with an Intel i7-12700k CPU, RTX 2080ti GPU, and 64 GB of RAM, in the PyTorch framework.

To ensure the fairness of the experiment, the following strategies on different methods were adopted. For the CNN, 20 HSI samples were used for training in each class. For HRWN, EndNet, CCR-Net, FGCN, and GM-VDFN, each class was trained using 20 HSI and LiDAR samples. For each method, Adam was used as the optimizer, the learning rate was set to 0.001, weight decay was 0.0005, and batch size was 100. In the comparison experiment, the parameter settings of each comparison method follow the original settings. To reduce the influence of randomness on the experimental results, we took the average of all the experimental results of 10 experiments. Three evaluation indexes were selected in the experiment, including overall accuracy (OA), accuracy of various classes (AA), and Kappa.

TABLE II
CLASSIFICATION PERFORMANCE ON HOUSTON 2013

Class	CNN	HRWN	EndNet	CCR-Net	FGCN	GLT-Net	GM-VDFN
Healthy grass	87.81	90.09	93.58	95.94	88.79	97.24	**97.64**
Stressed grass	87.28	87.84	88.25	92.95	89.79	96.60	**97.89**
Synthetic grass	71.20	97.34	99.11	99.70	**100**	**100**	98.67
Trees	93.87	93.30	98.04	95.18	93.14	**99.35**	94.20
Soil	99.10	99.59	**99.67**	99.18	99.10	99.26	98.45
Water	**98.03**	96.72	91.48	78.36	**98.03**	97.71	95.08
Residential	88.30	75.08	90.54	88.14	88.54	90.63	**92.23**
Commercial	78.59	63.32	81.45	72.88	**89.05**	83.42	77.94
Road	88.31	85.15	74.19	62.50	**89.29**	80.11	72.97
Highway	81.52	85.75	83.60	77.88	78.79	75.81	**91.30**
Railway	51.11	77.53	88.15	77.61	84.86	87.41	**97.78**
Parking lot 1	64.72	75.27	66.12	79.72	50.29	90.44	**91.10**
Parking lot 2	**95.32**	92.20	63.25	78.40	91.98	87.08	86.41
Tennis court	99.02	99.75	99.26	92.16	97.79	**100**	**100**
Running track	**100**	**100**	98.28	95.94	**100**	**100**	99.69
OA (%)	83.60	85.67	87.24	85.36	87.36	91.28	**92.04**
AA (%)	85.61	84.52	87.67	85.77	89.30	92.34	**92.76**
Kappa (%)	82.29	87.93	86.21	84.16	86.34	90.58	**91.40**

TABLE III
CLASSIFICATION PERFORMANCE ON TRENTO

Class	CNN	HRWN	EndNet	CCR-Net	FGCN	GLT-Net	GM-VDFN
Apple trees	94.29	98.68	98.75	97.56	91.31	97.91	**99.93**
Buildings	96.08	77.49	70.93	94.24	**99.55**	98.89	96.57
Ground	**99.56**	97.17	97.60	80.83	97.17	94.12	92.81
Woods	99.53	99.55	**100**	99.89	**100**	99.90	**100**
Vineyard	90.81	96.94	99.04	99.14	98.98	98.73	**100**
Roads	89.63	93.15	97.84	89.51	**98.18**	95.91	94.93
OA (%)	94.42	95.70	96.45	97.40	97.89	98.62	**99.02**
AA (%)	94.98	94.28	94.03	93.53	97.53	**97.58**	97.37
Kappa (%)	92.64	93.83	95.26	96.53	97.18	98.17	**98.69**

C. Parameter Analysis

1) Influence of Balance Coefficients: For simplicity, we take the Trento dataset as an example and set balance coefficients λ_1 and λ_2 to {0.001, 0.01, 0.1, 1, 10}, respectively. Fig. 4 demonstrates the variation of OA with different balance coefficient settings, from which can be seen the following.

1) With the increase of λ_1, OA shows an increasing and then decreasing trend. This is because: the small λ_1 limits the effectiveness of STC-GM, which makes it difficult to effectively constrain the model to suppress spectral noise during the learning process, impairing the model performance. The large λ_2 leads to an over-alignment of the spatial topology structure of HSI and LiDAR, which prevents the model from capturing complementary information.
2) With the increase of λ_2, OA shows an increasing and then decreasing trend, which is because: the small λ_2 limits the modeling ability of M-VAE on the multimodal joint distribution, resulting in the difficulty of suppressing the modal heterogeneity. The large λ_2 lets the model focus too much on the task of modeling the joint multimodal distribution and neglect the key classification task, resulting in limited model performance.
3) The highest OA was achieved by the model when $\lambda_1 = 0.1$ and $\lambda_2 = 1$, so we set λ_1 to 0.1 and λ_2 to 1.

2) Influence of GCN Layers Number: For simplicity, we take the Trento dataset as an example and set GCN layers number L to {1, 2, 3, 4, 5}, Fig. 5 demonstrates the variation of OA with different L, from which it can be seen that: With the increase of L, OA shows an increasing and then decreasing trend. This is because the small L limits the learning ability of the model, while large L leads to the problem of over-smoothing, where different classes of ground objects have similar feature representations. Therefore, we set L to 3 to strike a balance between model learning ability and over-smoothing.

D. Comparative Experiments

It can be observed from Tables II–V.

1) Compared to the CNN using only HSI data, the multimodal methods achieved higher classification accuracy. This is because LiDAR provides elevation information of ground objects, which can accurately describe the different heights of ground objects.

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TABLE IV
CLASSIFICATION PERFORMANCE ON MUUFL

Class	CNN	HRWN	EndNet	CCR-Net	FGCN	GLT-Net	GM-VDFN
Trees	80.01	90.63	76.68	76.63	**92.60**	84.64	87.01
Mostly grass	70.66	77.60	**79.32**	63.22	68.66	70.78	71.85
Mixed ground surface	51.85	56.34	54.58	**74.10**	58.64	64.09	62.20
Dirt and sand	78.68	32.34	84.94	74.03	84.77	82.00	**86.16**
Road	68.86	44.53	**86.16**	81.51	65.71	76.15	76.76
Water	**99.78**	83.86	98.21	97.76	87.44	98.88	98.42
Building shadow	69.36	64.26	72.80	71.62	83.42	**93.18**	83.54
Building	88.75	86.67	86.50	87.57	69.18	86.72	**92.42**
Sidewalk	61.25	**73.48**	46.37	64.03	52.82	57.44	58.79
Yellow curb	61.35	**93.25**	74.23	87.12	69.94	80.37	81.82
Cloth panels	89.56	95.18	93.57	93.17	97.19	**98.39**	95.80
OA (%)	74.47	75.46	75.97	76.79	78.51	79.82	**81.06**
AA (%)	74.56	68.20	77.58	79.16	75.49	81.15	**81.34**
Kappa (%)	67.64	72.56	69.82	70.80	71.92	74.32	**75.47**

TABLE V
CLASSIFICATION PERFORMANCE ON HOUSTON 2018

Class	CNN	HRWN	EndNet	CCR-Net	FGCN	GLT-Net	GM-VDFN
Health grass	88.99	**96.25**	92.23	93.58	90.94	94.63	90.93
Stress grass	83.86	88.28	86.94	**89.39**	88.71	88.12	87.67
Artificial turf	**100**	**100**	**100**	**100**	**100**	**100**	**100**
Evergreen trees	91.63	95.23	94.81	94.97	95.66	95.50	**96.50**
Deciduous trees	87.41	**97.00**	91.59	92.93	95.65	94.03	96.41
Bare earth	97.83	95.62	98.39	99.16	99.63	98.99	**99.82**
Water	**100**	98.96	**100**	**100**	**100**	**100**	98.44
Residential buildings	86.32	88.79	89.31	88.61	91.04	91.34	**93.39**
Non-residentialbuildings	85.75	85.26	88.94	88.49	86.91	**91.23**	89.75
Roads	37.12	61.68	52.97	57.82	59.02	57.70	**61.95**
Sidewalks	43.70	60.22	47.30	48.45	62.71	57.72	**65.30**
Crosswalks	71.91	60.37	82.01	83.59	87.15	**88.60**	88.14
Major thoroughfares	45.69	49.43	56.26	62.68	64.84	64.68	**67.50**
Highways	85.88	94.08	90.86	93.44	96.78	96.22	**96.94**
Railways	97.48	99.66	99.53	99.25	99.58	**99.75**	99.74
Paved parking lots	80.81	**95.03**	92.05	93.45	94.28	94.76	94.02
Unpaved parking lots	**100**	**100**	**100**	**100**	**100**	**100**	**100**
Cars	85.41	92.56	91.72	90.04	93.38	92.30	**95.82**
Trains	94.22	98.27	99.22	99.54	99.44	99.62	**99.68**
Stadium seats	99.31	98.78	99.79	99.80	99.81	99.73	**99.84**
OA (%)	75.33	80.25	80.58	81.71	82.56	84.04	**84.68**
AA (%)	83.17	87.77	87.70	88.14	90.28	90.25	**91.08**
Kappa (%)	68.96	75.20	75.34	76.77	77.92	79.61	**80.48**

2) Compared to HRWN, EndNet, and CCR-Net, FGCN demonstrates better classification performance. This is attributed to FGCN's utilization of graph convolution for feature extraction, which enables it to adapt to the irregular spatial topology structure of ground objects. By suppressing information interference from background and different ground objects in spatial neighborhoods, FGCN can accurately classify different classes of ground objects.
3) The GM-VDFN exhibits the best classification performance. This is because: First, the STC-GM makes the features of HSI and LiDAR have topological consistency in space by aligning the spatial distribution of the two modalities, which improves the intraclass consistency of features. Second, MDGAM dynamically explores personalized fusion patterns for different samples, which adapts to various spatial scales of ground objects. Third, M-VAE models the multimodal joint distribution of HSI and LiDAR, which effectively solves the challenge brought by heterogeneity to multimodal data fusion, and realizes the effective integration of multimodal information.

E. Comparative Experiments Under the Standard Settings

Considering that Houston 2013 and Trento datasets provide the standard training and test sets, we have used the standard training set and test sets of Houston 2013 and Trento datasets as shown in Table VI to compare the classification performance of the proposed method and its comparison methods, and the results are shown in Tables VII and VIII. It can be seen that GM-VDFN achieved the highest classification accuracy under the standard training set and test set settings of Houston 2013 and Trento datasets, verifying its effectiveness in the multimodal remote sensing data classification task.

TABLE VI
STANDARD TRAINING AND TEST SETS OF HOUSTON 2013 AND TRENTO DATASETS

Houston 2013				Trento			
	Class	Sample number			Class	Sample number	
No	Name	Training	Test	No	Name	Training	Test
1	Healthy grass	198	1053	1	Apple trees	129	3905
2	Stressed grass	190	1064	2	Buildings	125	2778
3	Synthetic grass	192	505	3	Ground	105	374
4	Trees	188	1056	4	Woods	154	8969
5	Soil	186	1056	5	Vineyard	184	10317
6	Water	182	143	6	Roads	122	3052
7	Residential	196	1072				
8	Commercial	191	1053				
9	Road	193	4059				
10	Highway	191	1036				
11	Railway	184	1054				
12	Parking lot 1	192	1042				
13	Parking lot 2	184	285				
14	Tennis court	181	247				
15	Running track	187	473				
	Total	2832	12197		Total	819	29395
Total (Training + Test)		15029		Total (Training + Test)		30214	

TABLE VII
OAS OF THE HOUSTON 2013 DATASET AND THE TRENTO DATASET

Dataset	CNN	HRWN	EndNet	CCR-Net	FGCN	GLT-Net	GM-VDFN
Houston 2013	85.13	87.93	88.75	89.83	90.71	93.82	**95.43**
Trento	92.11	96.80	97.43	97.95	98.13	98.95	**99.43**

TABLE VIII
KAPPAS OF THE HOUSTON 2013 DATASET AND THE TRENTO DATASET

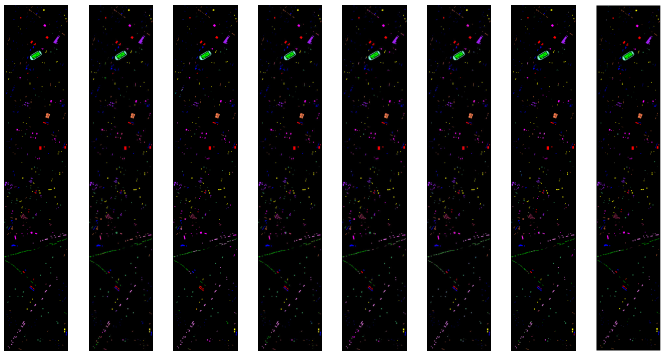
Dataset	CNN	HRWN	EndNet	CCR-Net	FGCN	GLT-Net	GM-VDFN
Houston 2013	84.67	86.71	87.26	88.73	89.86	94.23	**94.62**
Trento	89.51	95.73	97.09	97.43	97.84	98.36	**98.92**

F. Visual Analysis

1) Visual Analysis of Classification Maps: Figs. 6–8 present the classification maps of all methods on three datasets, from which it can be seen that the classification maps of GM-VDFN demonstrate the clearest classification boundaries. This is because: 1) GM-VDFN can capture the dependence relationship between ground objects in irregular spatial regions, effectively suppressing the interference of background and different-class ground objects; and 2) GM-VDFN mines modal information from multiple spatial scales, enabling the model to fully learn the spatial topology structure of ground objects. By forcing ground objects in the same spatial topology region to have similar feature representations, the spatial consistency of the classification result is improved.

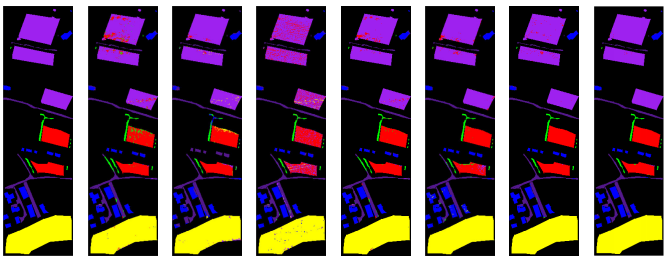
2) Visual Analysis: Fig. 9 displays the t-distributed stochastic neighbor embedding (t-SNE) of multimodal fusion classification methods on the Trento. It can be observed that GM-VDFN exhibits the best intraclass consistency and interclass separation. This is because, on the one hand, STC-GM can effectively suppress spectral noise by using spatial prior and accurate spatial topology information of LiDAR, which

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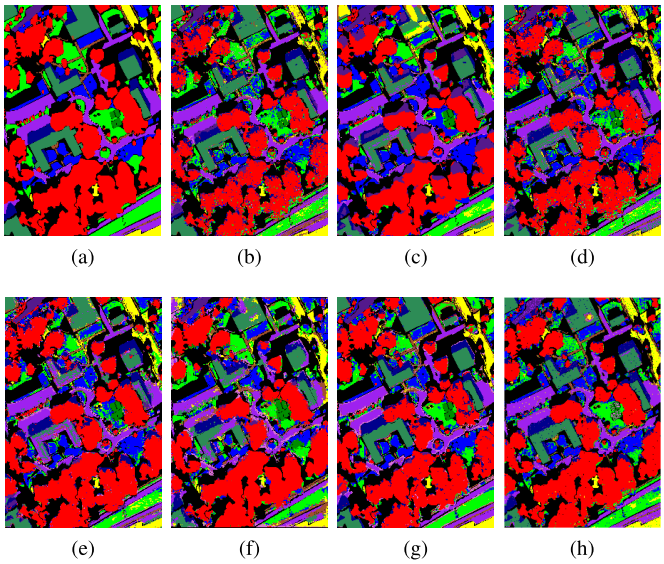
(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. Classification maps of all methods on Houston dataset. (a) Label map. (b) CNN. (c) HRWN. (d) EndNet. (e) CCR-Net. (f) FGCN. (g) GLT-Net. (h) GM-VDFN.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 7. Classification maps of all methods on Trento dataset. (a) Label map. (b) CNN. (c) HRWN. (d) EndNet. (e) CCR-Net. (f) FGCN. (g) GLT-Net. (h) GM-VDFN.



(a) (b) (c) (d)

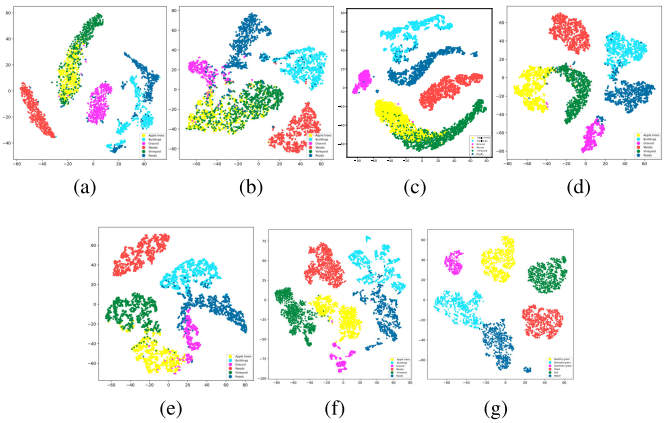
(e) (f) (g) (h)

Fig. 8. Classification maps of all methods on MUFFL dataset. (a) Label map. (b) CNN. (c) HRWN. (d) EndNet. (e) CCR-Net. (f) FGCN. (g) GLT-Net. (h) GM-VDFN.

reduces the intraclass variance of the feature. On the other hand, GM-VDFN fully excavates multiscale modal information and uses the complementary properties of HSIs and LiDAR to effectively separate different ground objects.

G. Ablation Study

To verify the contribution of different techniques of GM-VDFN to the ultimate classification result, we ablate



(a) (b) (c) (d)

(e) (f) (g)

Fig. 9. (a) CNN. (b) HRWN. (c) EndNet. (d) CCR-Net. (e) FGCN. (f) GLT-Net. (g) GM-VDFN.

TABLE IX

ABLATION EXPERIMENTS

	GM-VDFN1	GM-VDFN2	GM-VDFN3	GM-VDFN
MDGAM	✘	✔	✔	✔
STC-GM	✘	✘	✔	✔
M-VAE	✘	✘	✘	✔

TABLE X

IMPACT OF DIFFERENT TECHNIQUES ON OAS (%)

Dataset	GM-VDFN1	GM-VDFN2	GM-VDFN3	GM-VDFN
Houston 2013	88.15	89.57	91.02	**92.04**
Trento	95.56	97.12	98.21	**99.02**
MUUFL	76.13	77.52	78.45	**80.65**

MDGAM, STC-GM, M-VAE, as shown in Table IX, and the OAs of these ablation models are shown in Table X.

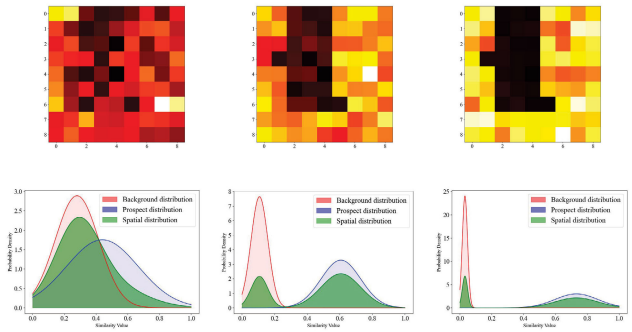
1) Effectiveness of MDGAM: Compared to GM-VDFN2, the OAs of GM-VDFN1 decreased by 1.42%, 1.56%, and 1.39% on three datasets. This indicates that MDDGM can fully exploit the modal information by exploring multiscale feature fusion patterns that are suitable for different ground objects.

2) Effectiveness of STC-GM: Compared with GM-VDFN3, which uses STC-GM to align the spatial distributions of two modalities, the OAs of GM-VDFN2 decreased by 1.45%, 1.09%, and 0.93% on three datasets, respectively. This indicates that STC-GM can effectively suppress spectral noise and enhance the discriminability of the model.

To show the effectiveness of STC-GM more clearly, we visualize the constraint effect of STC-GM, as shown in Fig. 10. From Fig. 10(a), in the absence of STC-GM, the ground objects in the same spatial topology region are affected by spectral noise and show poor consistency. From Fig. 11(b) and (c), the spatial distribution of HSIs and LiDAR is aligned with STC-GM, which results in a consistent feature representation of ground objects in the same spatial topology region of HSIs.

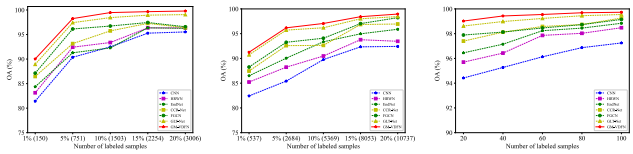
3) Effectiveness of M-VAE: Compared to GM-VDFN, which obtains multimodal fusion features using M-VAE, GM-VDFN3 has a decrease in OAs of 1.02%, 0.81%, and 2.20% on the three datasets, respectively. It is shown that the multimodal fusion features obtained by learning the joint distribution of GM-VDFN3 can explore the complex nonlinear

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(a) (b) (c)

Fig. 10. Visualization of STC-GM. (a) In the absence of STC-GM, (top) spatial topology matrix and (bottom) spatial distribution are calculated based on HSI features. (b) With STC-GM, (top) spatial topology matrix and (bottom) spatial distribution calculated based on HSI features are calculated. (c) (Top) Spatial topology matrix and (bottom) spatial distribution calculated based on LiDAR features.



(a) (b) (c)

Fig. 11. Classification performance of all methods with different training set allocation settings on the three datasets. (a) Houston 2013 dataset. (b) MUUFL dataset. (c) Trento dataset.

relationship between the modalities. This can better preserve the complementary features and provide more discriminative features for the ground object classification.

H. Influence of Training Set Allocation Settings

We compared the performance of GM-VDFN and its comparison methods under different training set allocation settings. For the Houston 2013 dataset and the MUUFL dataset, we gradually increased the number of training samples from 1% to 20%. For the Trento dataset, we gradually increased the number of training samples from 20 per class to 100 per class. The experimental results are shown in Fig. 11.

1) As the number of training samples increases, the OAs of all methods increase, which is because more training samples can better reflect the real data distribution and can help the model to better distinguish the classes of ground objects.
2) GM-VDFN exhibits the highest classification performance under different training set allocation settings, indicating its good adaptability to changes in sample number.

I. Computational Cost and Running Time

To show computational costs and running times of all methods, we took the Houston 2013 dataset as an example and counted their running time, floating-point operations (FLOPs), and number of parameters, as shown in Table XI. Among the all methods, GM-VDFN ranks fourth in terms of running time, fifth in terms of FLOPs, and fourth in terms of a number of parameters, and computational cost is in the middle. This is because the multiscale operation of GM-VDFN increases the scale of model inputs, which results in sacrificing computational cost.

TABLE XI
COMPUTATIONAL COST ON THE HOUSTON 2013 DATASET WITH DIFFERENT METHODS

Methods	CNN	HRWN	EndNet	CCR-Net	FGCN	GLT-Net	GM-VDFN
Running time (s)	384.25	165.31	92.42	1351.32	12135.12	743.24	658.25
FLOPs (M)	1.83	1.53	0.89	2.46	13.46	312.28	8.46
Parameters (M)	1.62	1.42	0.93	1.35	1.13	0.55	1.24

V. DISCUSSION ON THE PRACTICAL APPLICATION OF THE PROPOSED METHOD

GM-VDFN has good spectral noise suppression ability and multiscale context information mining ability, which is practical in the real scene. Take the detection of diseases and pests in wild crops as an example: On the one hand, due to the variability of weather, the model needs to face spectral noise interference induced by complex weather conditions [48], and GM-VDFN can use the spatial topology information of LiDAR to achieve precise suppression of such spectral noise. On the other hand, due to the unknown spatial distribution of pests and diseases in the real world, the model needs to adapt to the potential complex scale changes in the spatial distribution. GM-VDFN can achieve intelligent adaptation to these complex scale changes by dynamically fusing multiscale information. In summary, GM-VDFN can achieve precise detection of diseases and pests in wild crops by integrating multimodal remote sensing data for pixel-level classification of diseases and pests in wild crops.

VI. CONCLUSION

A novel fusion network is proposed, which effectively solves the three major challenges in HSI and LiDAR multimodal fusion classification tasks: Spectral noise suppression, full mining modal information, and multimodal data fusion. GM-VDFN has the following advantages.

1) By aligning the spatial distribution of HSIs and LiDAR, it can improve the intraclass consistency of ground features and reduce the performance damage caused by spectral noise to the model.
2) GM-VDFN captures shareable class identification information in multiscale features and fully mines contextual information by customizing personalized fusion models for different samples.
3) GM-VDFN can transform the problem of multimodal fusion into a multimodal joint distribution learning problem, overcoming the obstacles caused by heterogeneity between modalities and providing a new paradigm for multimodal data fusion. The experimental results show that GM-VDFN can achieve the most advanced classification performance on four public HSI datasets.

However, this article only focuses on processing HSIs and LiDAR data that have the same spatial resolution and are collected at the same time. It is difficult for our proposed method to use multisource remote sensing data with different spatial

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resolutions and multitemporal. The SpectralGPT [49] can capture shared knowledge from multisource remote sensing data with different attributes, providing a general foundation model for remote sensing data processing. Inspired by [49], in future work, we will try to combine the self-supervised learning and encoder for visible tokens to establish a collaborative learning mechanism. Through the above operation, the rich knowledge of multisource remote sensing data is integrated to improve the classification accuracy of the model.

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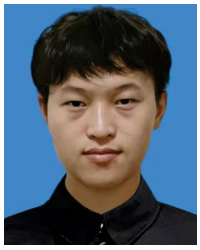
Haoyu Wang (Member, IEEE) received the M.S. and Ph.D. degrees from China University of Mining and Technology, Xuzhou, China, in 2017 and 2021, respectively.

He is currently a Post-Doctorate with the School of Information and Control Engineering, China University of Mining and Technology. His main research interests include hyperspectral image analysis, meta-learning, and transfer learning.



Xiaomin Liu (Member, IEEE) received the B.S. degree from the School of Information and Electrical Engineering, China University of Mining and Technology, Xuzhou, China, in 2014, and the Ph.D. degree from the School of Information and Control Engineering, China University of Mining and Technology, in 2021.

She is currently a Lecturer with the School of Information and Control Engineering, China University of Mining and Technology. Her research interests include reinforcement learning, machine learning, and image processing.



Zhenzhuang Qiao is currently pursuing the M.E. degree with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, China.

His research interests include image processing, few-shot learning, and domain generalization.



Guoqing Wang (Member, IEEE) received the B.S. degree from China University of Mining and Technology (CUMT), Xuzhou, China, in 2014, and the Ph.D. degree from Harbin Engineering University, Harbin, China, in 2019.

From October 2017 to May 2019, he was a Visiting Scholar with the Department of Electrical Engineering, Columbia University, New York, NY, USA. He is currently an Associate Professor with CUMT. His research interests include robust state estimation, information fusion, and their applications in navigation technology.



Haotian Chen is currently pursuing the M.E. degree with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, China.

His research interests include industrial data-driven modeling and machine learning.

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